

Loan Prediction Research

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```
import shap
import pandas as pd

import scipy.stats as stats
import joblib

# setting the working directory
import os
os.chdir("/Users/thomasnguyen/loan_dataset/")

## Importing the variables and models

xgboost_model = joblib.load("model/optimised_xgboost_model.pkl")
random_forest_model = joblib.load("model/optimised_random_forest_model.pkl")
logistic_regression_model = joblib.load("model/optimised_logistic_model.pkl")
df2 = pd.read_csv("data/cleaned_loan_dataset.csv")
df = pd.read_csv("data/loan_dataset.csv")
shap_values = joblib.load("data/random_forest_shap_values.pkl")
X_test = joblib.load("data/X_test.pkl")
```

Abstract

Bank Loans are

In this analysis, I used Python (v3.11.3), Pandas, Scikit-learn and SHAP to investigate a loan dataset to uncover the most important variables when rejecting or approving a loan.

Methods

Software

In this analysis, I used Python (v3.11.3) alongside the following packages:

- category_encoders==2.6.4
- joblib==1.4.2
- matplotlib==3.9.2
- numpy==2.1.3
- pandas==2.2.3
- plotly==5.23.0
- scikit_learn==1.5.2

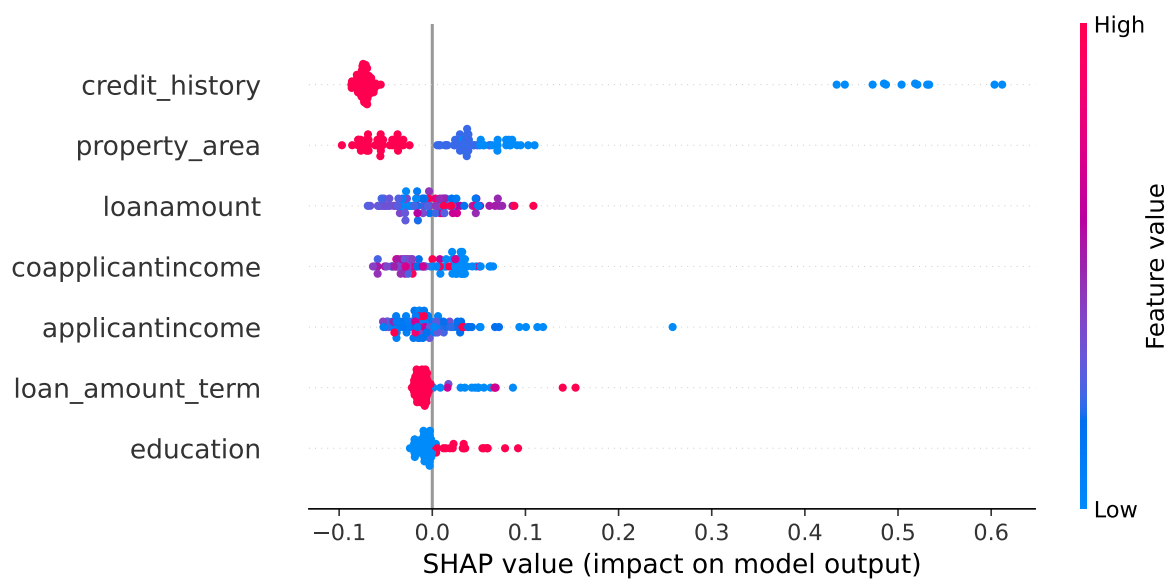
- scipy==1.14.1
- seaborn==0.13.2
- shap==0.45.1
- streamlit==1.37.0
- streamlit_shap==1.0.2

Data Availability

The data I used is available at: <https://www.kaggle.com/datasets/sonalisingh1411/loan-approval-prediction?select=Training+Dataset.csv>

Results

```
shap.plots.beeswarm(shap_values[:, :, 0])
```



Discussion

Conclusion

notes: - Semi-urban was encoded the highest, Rural is the lowest, urban is the middle - Married: 1 = Yes, 0 = No

1. All models agreed with each other
2. Loan amount, applicant income, co-applicant income, loan term inversely correlated with acceptance likelihood
3. males are slightly more likely to be accepted than females
4. Graduates are more likely to be accepted for a loan (education is important!)
5. dependents, gender, self-employed status and applicant income are the lowest predictors of loan status
6. Urban areas are the most likely to be accepted for a loan, followed by semi-urban and then rural properties
7. Married people are more likely to be granted a loan

Concordant papers

Limitations

1. Low sample size (many samples deleted)
2. Data was found on kaggle with no DOI citation